

Master thesis

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AURA: Adaptive, Understandable, and Reliable AI for kidney transplant aftercare

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Declaration on the Use of Generative AI Tools

Declaration (first page) is to be included in the submitted version of the student's report.

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AI tool name	Tool version, date of use	Purpose (according to Generative AI Usage Taxonomy, see next pages), additional remarks (if needed)
Chat-GPT	5.3, Until March 2026	Paraphrase and Reword, Grammar and spelling check, debugging
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I hereby declare, that I am the sole author and composer of my thesis and that no other sources or learning aids, other than those listed, have been used. Furthermore, I declare that I have acknowledged the work of others by providing detailed references of said work. I also hereby declare that my thesis has not been prepared for another examination or assignment, either in its entirety or excerpts thereof.

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Abstract

Kidney transplant recipients require intensive long-term after-care to mitigate the risk of graft rejection, graft loss, and mortality. The Temporal Fusion Nexus (TFN) has demonstrated state-of-the-art performance on these prediction tasks, yet its suitability for clinical deployment depends on more than predictive accuracy alone. This thesis introduces AURA, a framework for evaluating multimodal clinical AI models in terms of understandability and reliability.

We apply and compare four explainability techniques — Integrated Gradients, TIMING, SHAP, and Concept Relevance Propagation — at both local and global levels on TFN, and evaluate their clinical alignment against a new expert study involving 20 nephrologists. Attribution quality is quantified using Normalised Discounted Cumulative Gain, with IG and TIMING achieving scores of 0.85 and SHAP achieving 0.96 for graft loss prediction. We further quantify TFN’s predictive uncertainty using Adaptive Conformal Inference and Risk-Controlling Prediction Sets, demonstrating mean feature coverage exceeding 90% across all features.

Our analysis reveals that TFN’s temporal decay mechanism causes the model to concentrate almost entirely on the most recent timesteps, which limits the utility of temporally-aware attribution methods.

We observed discrepancies with clinical expert opinion. Importantly, we show that these discrepancies are medically justifiable rather than indicative of model failure, providing a framework for interpreting misalignment between model behaviour and clinical knowledge.

AURA demonstrates that strong benchmark performance and clinical trustworthiness are complementary and measurable properties, and provides a reusable evaluation methodology for future multimodal clinical AI systems.

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1 Introduction

Despite constant medical advancements, recipients of a kidney transplant (KTx) remain at risk of severe outcomes, including graft rejection, graft loss, and mortality [1]. Ensuring patient safety requires intensive aftercare to enable timely clinical intervention. To support the already stretched medical workforce, researchers worldwide are developing AI-driven clinical decision support systems capable of predicting adverse events. However, due to the complexity and multimodality of data involved in KTx aftercare, conventional approaches struggle to reliably predict outcomes such as graft loss [2]. Our group previously proposed the multi-modal architecture *Temporal Fusion Nexus* (TFN) as a solution to this problem, demonstrating state-of-the-art predictive performance [3].

Strong benchmark results alone, however, are insufficient for clinical adoption. Physicians must be able to understand and scrutinise a model’s reasoning before integrating it into their workflow, making explainability a critical requirement [4]. Equally, a system deployed in a life-critical setting must be reliable and well-calibrated, so that its predictions neither mislead clinicians nor cause unnecessary harm [5]. This work therefore introduces *AURA* — a framework to evaluate if multi-modal models for KTx aftercare are **A**daptive, **U**nderstandable, and **R**eliable **AI**, — to systematically evaluate and extend TFN along these dimensions.

Specifically, this thesis addresses the following research questions:

1. How can the interpretability of a multi-modal model be addressed on both global and local levels?
2. How well do the model’s learned representations and feature importances align with clinical expert knowledge, and what do observed discrepancies reveal?
3. How can predictive uncertainty be quantified for TFN, and how could this serve as a basis for improving reliability and personalisation?

The contributions of this thesis are as follows:

- **Explainability analysis.** We apply and compare multiple attribution techniques — including Integrated Gradients, TIMING, SHAP, and CRP — at both local and global levels on TFN, providing new insights into the model’s decision-making and validating its clinical alignment.
- **Uncertainty quantification.** We evaluate TFN’s predictive uncertainty using ACI and RCPS, demonstrating that the model produces well-calibrated, confident predictions, and outline how the resulting coverage metrics could inform future personalisation strategies.
- **Expert study.** We conduct a structured study with expert nephrologists, eliciting their prioritised feature rankings across three clinically challeng-

ing prediction tasks. The resulting ratings serve as an evaluation benchmark for attribution quality and contribute a resource likely to be valuable for future research.

The remainder of this thesis is structured as follows. Section 2 introduces the clinical and technical background. Section 4 describes the methodology. Sections 6 and 7 present and discuss the experimental findings and outline directions for future research.

2 Theoretical Background

Processing electronic health records (EHRs) presents complex challenges. EHRs naturally comprise multiple data modalities, including structured data such as vital signs and laboratory measurements [6], as well as unstructured free text such as notes and reports [7]. Bringing these sources together for coherent analysis is difficult [8]. Nevertheless, addressing this challenge is important, as vital signs alone do not provide sufficient context to understand a patient’s case fully, and clinicians rely heavily on written reports [9]. The information they contain is therefore highly relevant to our model. The inherently different data types suggest that a solution may be found in multimodal modelling. This, however, introduces a further set of problems: temporal data in EHRs are often sampled irregularly, contain missing information, and span long timeframes, all of which pose substantial challenges for classical LSTMs [10].

To address these issues, our group previously proposed the Temporal Fusion Nexus model (TFN) [3]. TFN is a multimodal, task-agnostic embedding model designed to combine irregularly sampled temporal data with unstructured medical texts. The model’s internal mechanisms and evaluation are briefly explained in the following paragraphs. For a detailed overview of the training process, see [3].

2.1 TM-LSTM

The temporal input of TFN is handled by a TM-LSTM, a version of T-LSTM [11] extended with a masking mechanism. T-LSTMs were developed to handle irregular time-series data, making them a strong basis for the challenges outlined above. By splitting the cell memory of the previous time step C_{t-1} into short- and long-term memory C_{t-1}^S and C_{t-1}^T , and by tracking the elapsed time between two time steps in a variable Δt , the use of a *decay* function allows the model to capture the fading relevance of older events [3].

$$C_{t-1}^S = \tanh(W_{decomp} * C_{t-1} + b_{decomp}) \quad (1)$$

$$C_{t-1}^T = C_{t-1} - C_{t-1}^S \quad (2)$$

$$C_{t-1}^* = C_{t-1}^S + C_{t-1}^T \odot e^{W_{decay} * \Delta t + b_{decay}} \quad (3)$$

The values of W_{decay} and b_{decay} are learned during training. Since our data are not only irregularly sampled but also sparse at times, a masking mechanism was added to handle missing data. A mask $M \in (0, 1)^{B \times T \times F}$, where B is the batch size, T the sequence length, and F the number of features, indicates whether a feature at a given time step is observed or missing, allowing the LSTM to distinguish between the two cases [3] without losing potentially important information encoded in the pattern of missingness [10]. Finally, TFN leverages a multi-head self-attention mechanism to capture the importance of different time points from the TM-LSTM output. A causal mask is applied to this attention mechanism to ensure that predictions at time t_n depend only on steps $t_0, \dots, t_{n-1}$. The patient's static data are processed through a static encoder, which formats them so that they can be concatenated with the time-series data and handled by the TM-LSTM [3].

2.2 Med-gte-hybrid

The other half of TFN, responsible for handling unstructured clinical notes, is based on Med-gte-hybrid [12], a language model derived from the BERT-based sentence transformer gte-large [13, 14] with the goal of extracting medical information from clinical narratives. Because NephroCAGE 3 was collected at a German clinic and therefore contains mostly German notes, Med-gte was fine-tuned on 20,000 randomly sampled German clinical sentences in a two-stage approach [3].

2.3 Multi-modal fusion via cross-attention

The outputs of the two previously described models are fused through cross-attention. Queries Q are computed from the TM-LSTM hidden layers, while keys K and values V result from a linear projection of the embedding space generated by Med-gte-hybrid. The output X produced by this attention mechanism is passed through a feed-forward layer that applies the same function learned when fusing the static and temporal data for the TM-LSTM. The result of this forward pass is a shared multimodal embedding space, termed the *Nexus* [3]. Up to this point, TFN remains task-agnostic. To solve a specific downstream task, for example estimating the likelihood of a patient experiencing graft loss within the next month, a multilayer perceptron is trained to solve that task using the Nexus as input. Figure 2 shows the overall architecture.

2.4 SHAP

SHapley Additive exPlanation Values [15] are a post-hoc local explainability technique that leverages the concept of Shapley values [16] to approximate

model attributions. Originally developed in cooperative game theory, Shapley values provide a principled way to fairly distribute a payout among players based on their individual contributions. In the context of explainability, the model is the game, the features are the players, and the attribution scores are the payout [17]. SHAP approximates Shapley values by repeatedly evaluating the model on different subsets of features, measuring how much each feature changes the model output when it is included or excluded. The final attribution for feature x_i is a weighted average of these marginal contributions across all possible feature subsets. In practice, computing exact Shapley values is exponentially expensive in the number of features, so SHAP is often approximated. The original paper provides a sampling-based approximation called KernelSHAP [15] that was also used during the past work on TFN [3] - though sampling-based approximations are increasingly being superseded by faster alternatives [17]. The previous SHAP analysis showed great alignment between TFN and the opinion of an expert nephrologist with over three decades of experience in the field, as shown in Fig 1.

3 Dataset

We use the same NephroCAGE subset as in [3]. NephroCAGE [18] is a collection of EHRs from kidney transplant recipients who received care at Charité Berlin between 2000 and 2020. Our subset includes 3,382 patients, 62% of whom are male and 38% female, with a mean age of 54 years. The dataset exhibits the typical challenges of EHRs described in Section 2. Specifically, we use patient demographic information as well as donor information, which we refer to as *static data*. In addition, each patient has *time-series data* in the form of laboratory measurements, which in most cases were taken irregularly over a long period of time. Finally, the dataset also contains notes written by clinicians after a patient’s assessment. Figure 3 shows an example of a patient record in NephroCAGE. To provide a clearer understanding of how such a record appears in practice, Table 1 presents the same patient’s data at time step t_0 , i.e. on the day of the transplant.

Blood pressure	Gender	Heart rate	Temperature	...	eGFR	...	Notes
-0.7072	m	-0.6565	-0.5390	...	-0.9114	...	"Anamnese: ..."

Table 1: Example illustrating what the data shown in Figure 3 represent in practice.

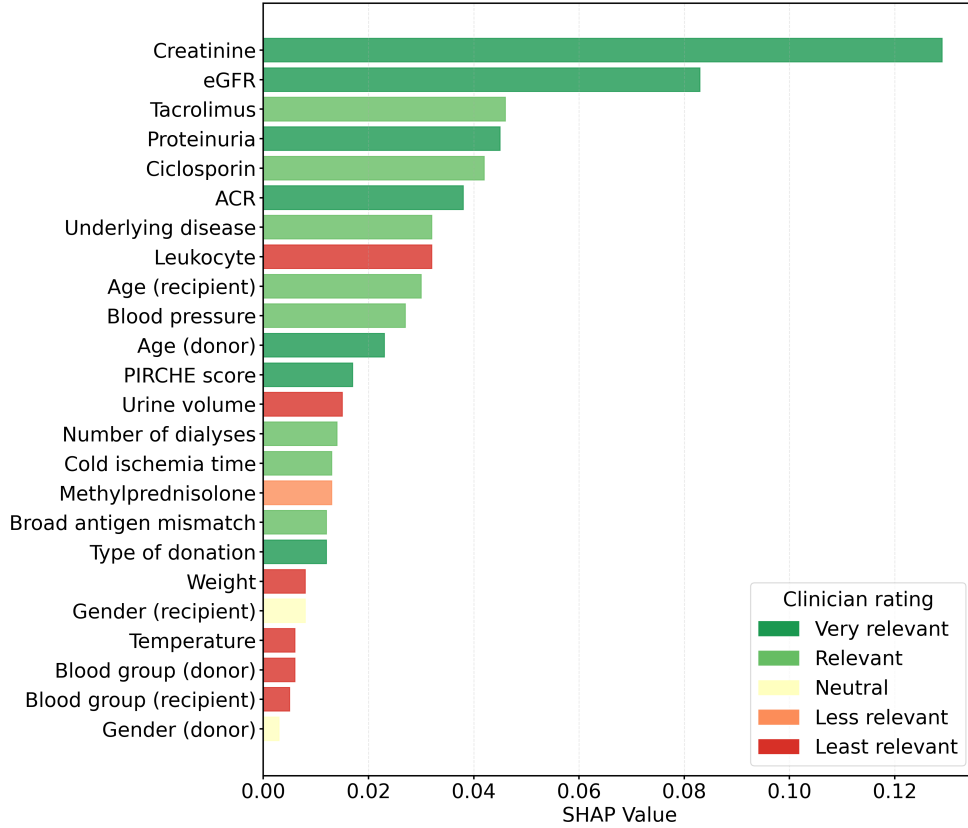


Figure 1: Mean absolute SHAP values for each feature, indicating their relative importance in TFN predictions for graft loss. The relevance rating of a clinical expert of each feature is shown too. Kidney function markers (creatinine and eGFR) appear as top features in SHAP-based feature importance, followed by immunosuppressant levels and indicators of kidney damage (proteinuria, ACR). The moderate correlation ($\rho = 0.62$) between SHAP values and clinician ratings demonstrate substantial alignment between the model’s learnt feature importance and clinical expertise [3].

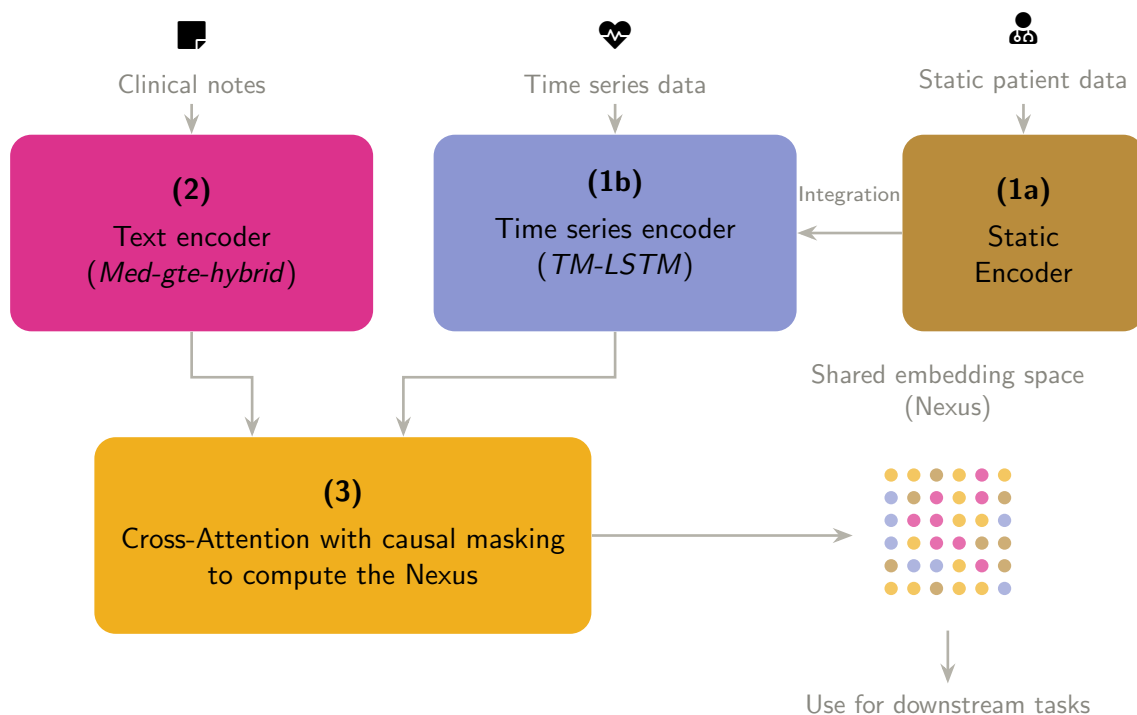


Figure 2: **The architecture of TFN.** Structured temporal data gets processed by a TM-LSTM (1), unstructured notes by the language model Med-gte-hybrid (2). Their outputs are fused via cross-attention (3), transformed a feed-forward layer into the Nexus. Can now be passed into an mlp solving the actual task (e.g. graft loss prediction) [3]

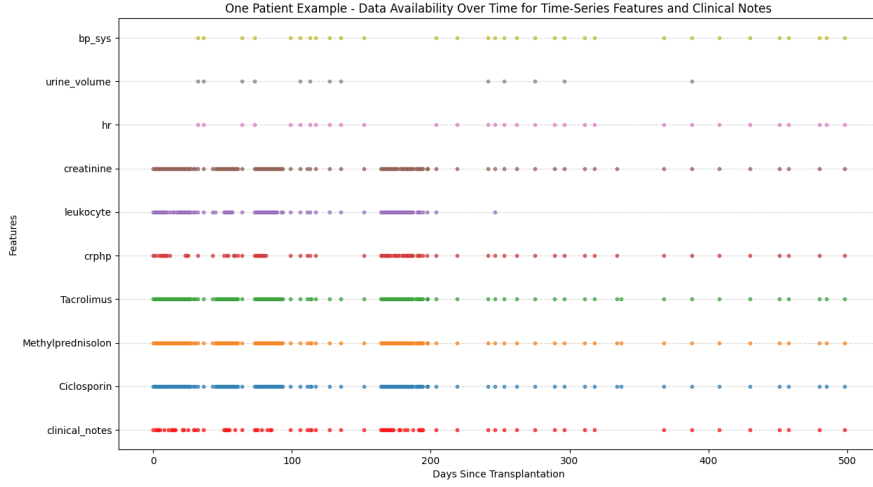


Figure 3: Trajectory of a representative KTx recipient in our cohort. The horizon axis shows relative time in days after transplantation. Static features include age, gender, and blood group. Longitudinally collected time-series features include blood pressure, urine volume, heart rate, serum creatinine, and related variables. Medications and clinical notes, including discharge summaries and radiology reports, were also documented longitudinally. This patient experienced two clinical outcomes: graft rejection (Day 163) and graft loss (Day 341) [3].

4 Methods

4.1 Integrated Gradients

Like SHAP (2), Integrated Gradients (IG) [19] is a post-hoc local explainability technique that is theoretically grounded in Shapley values. The main difference between the two methods lies in how they calculate feature attributions. Most modern attribution methods fall into two categories: approximation-based methods such as SHAP and gradient-based methods such as IG [20].

Given a DNN F with input x , IG computes an attribution A for features $x_i \in x$ with respect to a baseline x' .

$$A(x, x') = (a_i, \dots, a_i) \in \mathbb{R} \quad (4)$$

The aforementioned baseline is defined as *the absence of a feature*, although this definition has been controversial [21]. For time series, the baseline is most often defined as zero. Theoretically, this is not equivalent to an absent feature, but in practice it works well in many cases. Once a baseline has been chosen, a linear path from x' to x is constructed. At each step along this path, the gradient $\frac{dF(x)}{dx_i}$ is computed. IG is then obtained by accumulating these gradients; in

other words, IG is the path integral along the path from x' to x .

$$IG_i(x) ::= (x_i - x'_i) \times \int_{\alpha=0}^1 \frac{dF(x' + \alpha \times (x - x'))}{dx_i} d\alpha \quad (5)$$

Since computing this integral exactly would be expensive, it is approximated using a Riemann integral, as proposed by the original authors [19].

$$IG_i^{approx}(x) ::= (x_i - x'_i) \times \sum_{k=1}^m \frac{dF(x' + \frac{k}{m} \times (x - x'))}{dx_i} \frac{1}{m} \quad (6)$$

Instead of integrating along the entire path, a fixed number of steps m is defined along the path. In most cases, values of m between 20 and 300 yield adequate results [19]. Figure 4 shows an example of what such a path might look like for time-series data. Arguably, the biggest contribution of IG is that its authors defined five axioms they considered essential for a reliable attribution method and then built IG to satisfy them. For our work, two of these axioms are particularly noteworthy:

- **Axiom 2: Implementation Invariance** Two networks are functionally equivalent if their outputs are equal for all inputs, despite having very different implementations. Attribution methods should satisfy implementation invariance, meaning that the attributions are identical for two functionally equivalent networks [19].
- **Axiom 3: Completeness** Integrated gradients satisfy an axiom called completeness, meaning that the attributions sum to the difference between the output of F at the input x and the baseline x' [19].

Completeness can be used both to fine-tune the number of steps m and as a sanity check for an implementation.

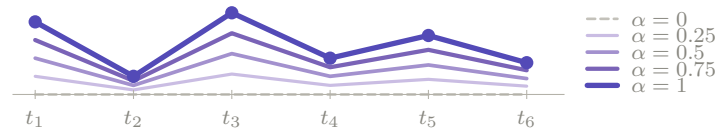


Figure 4: Interpolations $x' + \alpha(x - x')$ for $\alpha \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$. The signal fades from grey ($\alpha = 0$, flat zero baseline) to deep purple ($\alpha = 1$, corresponding to the input x).

4.2 TIMING

While SHAP and IG have proven themselves to be strong methods, in theory they are not optimal for analysing TFNs because neither method can explicitly capture temporal relationships. To improve IG in this regard, Jang et al.

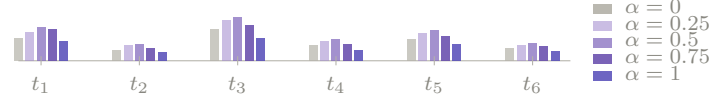


Figure 5: Gradients ∇F at each α step for a 6-timestep input. Each cluster shows five bars, one per α value (light grey $\rightarrow$ deep purple). Bar height encodes gradient magnitude.



Figure 6: Final attribution $IG_i = (x_i - x'_i) \cdot \int_0^1 \frac{\partial F}{\partial x_i} d\alpha$ using Gradients ∇F at each α step seen in Fig 5.

[22] proposed *Time Series Integrated Gradients* (TIMING), a post-hoc local explainability method based on IG that has been shown to capture temporal relationships while achieving state-of-the-art performance. To achieve this, at every step m_i along the path, the data are split into multiple segments. A randomly generated mask is applied to each segment. The masked points are not interpolated; instead, they retain their original value from x . Because this mask is regenerated at every step, the process creates a highly fluctuating path from x' to x . This fluctuation makes it possible to observe how masking different features changes $F(x)$. Because TIMING applies masking at the segment level, it can reflect temporal dependencies in each IG computation [22]. Given a binary mask $M \in \{0, 1\}^{T \times D}$ and a time-series input $x_{t,d}$, where T is the number of time steps, D is the feature dimension, and t and d index those respective sets, MaskingIG creates a baseline by applying the mask M to x via point-wise multiplication:

$$\text{MaskingIG}_{t,d}(x, M) = x_{t,d} M_{t,d} \times \int_{\alpha=0}^1 \frac{dF(\alpha(M \odot x) + (1 - M) \odot x)}{dx_{t,d}} d\alpha \quad (7)$$

TIMING then divides x into a specified number of segments n , each with a length between s_{min} and s_{max} . A mask generator G then calculates the masks used with MaskingIG for each segment. Accumulating the MaskingIG results across all segments yields the final attribution. Figure 7 shows an example of how masking changes the interpolation path.

$$\begin{aligned} \text{TIMING}_{t,d}(x, n, s_{min}, s_{max}) \\ = \mathbb{E}_{m \sim G(n, s_{min}, s_{max})} [\text{MaskingIG}_{t,d}(x, M) | M_{t,d} = 1] \end{aligned} \quad (8)$$

While TIMING preserves most of IG’s axiomatic guarantees, the masking procedure unfortunately breaks completeness.

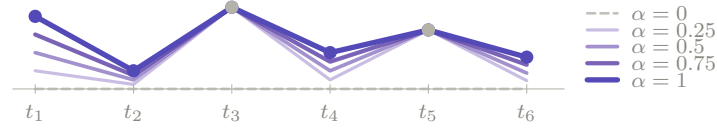


Figure 7: Interpolations $x' + \alpha(x - x')$ for $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$. The signal fades from grey ($\alpha = 0$, flat zero baseline) to deep purple ($\alpha = 1$, corresponding to the input x), except at t_3 and t_5 , where values are randomly masked.

4.3 Concept Relevance Propagation

While the aforementioned local explainability methods provide insight into *why* a model makes a particular decision, in the sense of which features contribute to it, they do not help to explain *how* a model arrives at that decision, for example what happens inside the layers or which relationships between features the model has learned. For this purpose, so-called global explainability methods are required. In this work, we chose Concept Relevance Propagation (CRP) [23].

CRP is an extension of Layer-wise Relevance Propagation (LRP) [24], which is itself a local explainability technique. Again assuming a DNN F with input x and L layers, such that $F(x) = f_L, \dots, f_1(x)$, LRP tracks all neuron activations during the forward pass and later ranks relevance by traversing the model in reverse. For every output j , it evaluates which corresponding inputs i contributed most to that output.

CRP extends this process by adding a set of conditions θ to LRP. A condition $c \in \theta$ could, for example, be *only track the relevance of inputs i that lead to output j in layer l* or *only track outputs j that are relevant to a final model prediction y* . Adding such conditions disentangles the relevance flows associated with concepts learned by the model, thereby allowing the computation of concept-conditional relevance mappings $R(x|\theta)$ [23]. To illustrate this, imagine F as an image classifier and x as an image of a dog. If the conditions are defined as $\theta = \{\text{dog}^L\}$, CRP outputs the relevance of the features leading to the prediction of the label *dog* in the model’s final layer; this would be functionally equivalent to LRP. However, CRP also allows more complex condition sets. Given $\theta = \{\text{dog}^L, \text{fur}^l\}$, it tracks the attributions for the dog image only in channels responsible for the representation of fur. Varying these conditions manually or automatically across different concepts makes it possible to identify which of those concepts have the highest relevance to a model’s prediction. To form a better understanding of *how* the model encodes these concepts, [23] propose a technique called Relevance Maximisation (RelMax). Given conditions θ encompassing a concept and a dataset D , RelMax computes $R(x|\theta)$ for every $x \in D$ and returns the top- k inputs x that yield the highest relevance under θ . In the previous example, RelMax would output pictures of dogs with

the highest relevance for the concept *fur*.

4.4 Conformal Prediction

First introduced in 1999 and further refined in 2005 by Vovk and Shafer [25][26], Conformal Prediction (CP) has established itself as a state-of-the-art method for quantifying the uncertainty of machine learning models and continues to be improved through ongoing research [27][28][29][30][31]. More precisely, due to computational limitations, machine learning research generally focuses on so-called *Split Conformal Prediction* [27][32]. Suppose a dataset D contains n data points (x, y) and a user-chosen error rate $\alpha \in (0, 1)$. In split CP, D is divided into a training set D_{train} and a calibration set D_{cal} . A model $F : x \rightarrow y$ is trained using D_{train} , after which a *nonconformity* score function $S : X \times Y \rightarrow \mathbb{R}_{\geq 0}$ calculates, for every sample in D_{cal} , the distance between the true label y and the prediction $F(x)$. A higher score indicates that the values conform *less* well to one another. From the resulting scores $s_1, \dots, s_n$, the $\frac{[(n+1)(1-\alpha)]}{n}$ quantile $\hat{q}$ is selected. This is the key quantity used to define a *valid* prediction set $C(x)$, where $C(x) \subset Y$ contains probable values for x and $C(x) := \{y : s(x, y) \leq \hat{q}\}$. Such a prediction set is called *valid* if and only if $P(Y_{test} \in C(X_{test})) \geq 1 - \alpha$ [32]. Fortunately, this guarantee holds whenever the input data in D_{train} and D_{cal} are *exchangeable* [26, Equation (2.2)], that is, when all permutations of the input yield the same result. In regression tasks, C would be a set of prediction intervals, which could be defined as follows, assuming an uncertainty-measuring function $u(x)$: $C(x) = [f(x) - u(x)\hat{q}, f(x) + u(x)\hat{q}]$ [32]. A sample (x_i, y_i) is called *covered* if the true value y_i lies within the prediction interval $C(x_i)$. The model F may then be described as *confident* if its prediction $\hat{y}$ lies within $C(x)$ at least $100 \times (1 - \alpha)\%$ of the time.

Unfortunately, classical CP is not a good measure for tasks involving time-series data, because the exchangeability assumption is violated. Permuting vital sign values from patients in NephroCAGE would have drastic effects on the predictions of a TFN. Therefore, an alternative to CP that can process temporal relationships is needed. We ultimately used Adaptive Conformal Inference (ACI) [28] because it still shows state-of-the-art performance when compared with newer methods [27], while also being comparatively simple to set up and model-agnostic. ACI belongs to the family of online CP algorithms [27][28], meaning that at each new time step $t \in \mathbb{N}$, the new information (x_t, y_t) is used to calculate an adapted prediction set C_t . To achieve this, in addition to α , ACI defines at every time step t a new value α_t , which is then used to define a prediction set $C_t(\alpha_t) := \{y : s_t(X_t, y) \leq \hat{q}_t(1 - \alpha_t)\}$, with X_t denoting the set of all observed values $x_1, \dots, x_t$. To adapt α_{t+1} , an error function is needed;

ACI measures error as miscoverage, i.e.

$$err_t := \begin{cases} 1, & \text{if } Y_t \notin C_t(\alpha_t) \\ 0, & \text{otherwise} \end{cases} \quad (9)$$

Fixing a parameter $\gamma > 0$ for control, the online update function can now be defined as follows:

$$\alpha_{t+1} := \alpha_t + \gamma(\alpha - err_t) \quad (10)$$

In [28, Appendix A], Gibbs & Candés provide various proofs supporting the mathematical soundness of their algorithm. Fig 8 shows an example of a prediction interval in time-series forecasting.

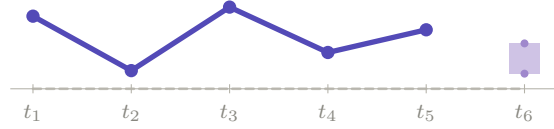


Figure 8: An example time-series. Assume a model F tries to predict the value at timestep t_6 . A good prediction should fall between the bounds displayed by the brightened dots with a probability of at least $(1 - \alpha) * 100\%$.

4.5 Risk-controlling Prediction Sets

In the same spirit as split CP, Bates et al. developed *Risk-controlling Prediction Sets* (RCPS) [33] as a new approach to calculating prediction sets for quantifying uncertainty. The main benefits of their technique over CP are that it can be used for a wider range of tasks, because it does not require the data to be *exchangeable*, and that it is compatible with various loss functions rather than simply tracking (mis)coverage, resulting in a more flexible method [33].

In 4.4, it was briefly explained that a prediction set $C(x)$ is always considered valid under the basic assumptions of CP; this is also called unconditional validity. Vovk later noted that, in certain high-risk use cases, it could be beneficial instead to define conditionally valid prediction sets [34], i.e. an entry in this set must not only reach a minimum conformity, but also a minimum significance measured by a function $p(y)$ that is higher than a threshold $\epsilon \in (0, 1)$: $C(x) := \{y : s(x, y) \leq \hat{q} \wedge p(y) \geq \epsilon\}$. In RCPS this idea was developed further, to directly encode the risk of a prediction into such a condition. Like in 4.4, Assume a divided dataset D , a model F trained using D_{train} , and a function mapping inputs x to prediction sets $\mathcal{Y} C : X \rightarrow \mathcal{Y}$ called predictor. Additional to the error threshold α , the user also defines a risk threshold $\delta \in (0, 1)$. Instead of CPs score function, assume a Loss function $L : Y \times \mathcal{Y} \rightarrow \mathbb{R}$, that encodes the consequence of an incorrect prediction. Now the risk of a function C can be defined as $R(C) := \mathbb{E}[L(y, C(x))]$. **C can be called a (α, δ) -risk-controlling**

prediction set, if $R(C) \leq \delta$ with a probability of at least $1 - \alpha$ [33].

To find a well-calibrated RCPS for a given task, assume a *nested* collection of predictors indexed by a variable $\lambda \in \Lambda$, i.e. $C_{\lambda_1} \subseteq C_{\lambda_2}$. After calculating a pointwise upper confidence bound $\hat{R}(C_\lambda)$ for each λ , define $\hat{\lambda}$ as the largest value in Λ that satisfies the condition $\hat{R}(C_\lambda) \leq \delta$. Then, with probability at least $1 - \alpha$, $C_{\hat{\lambda}}$ has risk $R(C_{\hat{\lambda}}) \leq \delta$, making $C_{\hat{\lambda}}$ an (α, δ) -RCPS, as proven in [33, Appendix A].

5 Study

During the original evaluation of TFN, a study with an expert nephrologist was conducted. For each of the three downstream tasks explained in 2, the clinician ranked all twenty-five features available in NephroCAGE on a 5-point Likert scale, with 1 indicating that a feature was irrelevant and 5 indicating that it was very relevant for the given task [3]. While insightful, the risk of bias in a study with $n = 1$ is high, so we designed a new study to obtain more robust results. There were two problems with the old survey format that had to be addressed. First, during test runs, multiple clinicians reported that the survey was difficult and tiring to complete, as they had to individually rank the importance of twenty-five features three times, even though they do not usually think about these features as isolated factors in their day-to-day work. Second, due to the expected small sample size, caused by the fact that there are not hundreds of nephrologists readily available, there was a considerable risk of a ceiling effect [35], which could have prevented the results from reflecting the actual top- k most important features for each task. An alternative to Likert scales is so-called Best Worst Scaling (BWS) [36], in which participants can rank only at the extremes by identifying an option as either the best or the worst. There is strong evidence that BWS surveys are easier to answer [37] and achieve better discrimination among attributes [38]. Leveraging these properties to address the aforementioned problems, a combination of BWS and Likert scale questions led to the following study design. For each of the three downstream tasks, the survey contains three questions. Q_1 asks the clinician to select the five most important (best) features, under the assumption that the decision over a patient would have to be based solely on those five metrics. Q_2 then presents a list of all features not selected in Q_1 and asks the expert which five additional features they would prefer to know in order to strengthen their decision. Q_3 then presents the remaining (worst) features on a 2-point Likert scale, on which the nephrologists can indicate whether they deem the feature irrelevant to the task or at least slightly relevant. At the end of each task, there was an opportunity to add remarks. In total, we recruited $n = 20$ nephrologists, although not all of them managed to complete the survey in its entirety. There are twenty responses for the task of graft loss, seventeen

for graft rejection, and sixteen for mortality, with experience levels ranging from one to thirty-five years. To determine the relevance of a feature, we first compute the percentage of responses placing it in the top 5, top 10, and little relevant categories, respectively. From these percentages, a simple weighted score is calculated by multiplying each result by a factor i and then summing the products. Irrelevant features do not add to the score.

$$i := \begin{cases} 1, & \text{if answer in top-5} \\ 0.66, & \text{if answer in top-10} \\ 0.33, & \text{else} \end{cases} \quad (11)$$

To keep the results comparable to the old format, and to facilitate their use in evaluating explainability results, quantiles are used to categorise the features again into five levels of relevance. Table 2 shows the resulting scores. Tables also including the computed relevance label can be found in the Appendix.

6 Results

Because all applied methods are post hoc and no internal changes were made to the model, a planned comprehensive re-evaluation of TFN was omitted.

6.1 Explainability Analysis

To better understand how TFN operates on a per-patient basis, the attribution analysis introduced in Section 2 was extended with IG and TIMING, and compared against the updated expert study. Both methods were applied to all three tasks over the full test set of 677 patients. All results of IG were verified via the completeness axiom.

To quantify the clinical alignment of feature attributions, we employ *Normalised Discounted Cumulative Gain* (NDCG) [39, 40], a metric originally developed for information retrieval that rates a ranking by a score $\in (0, 1)$, rewarding relevant entries appearing at the top. A score of 1 indicates a perfectly sorted ranking.

For the task of graft loss prediction at a 30-day horizon, both IG and TIMING achieved an NDCG score of 0.85, reflecting strong alignment with clinical opinion. The primary source of error for both methods was the high importance attributed to systolic blood pressure, a feature rated as less relevant by the clinical experts, whilst both correctly identified creatinine, eGFR, and urine volume as the most important features. The full results are shown in Figures 9–10. The earlier SHAP attributions achieved a near-perfect NDCG score of 0.96. However, their Spearman correlation with the expert rankings dropped from $\rho = 0.62$ under the previous study to $\rho = 0.34$ under the new one, though this still indicates moderate alignment.

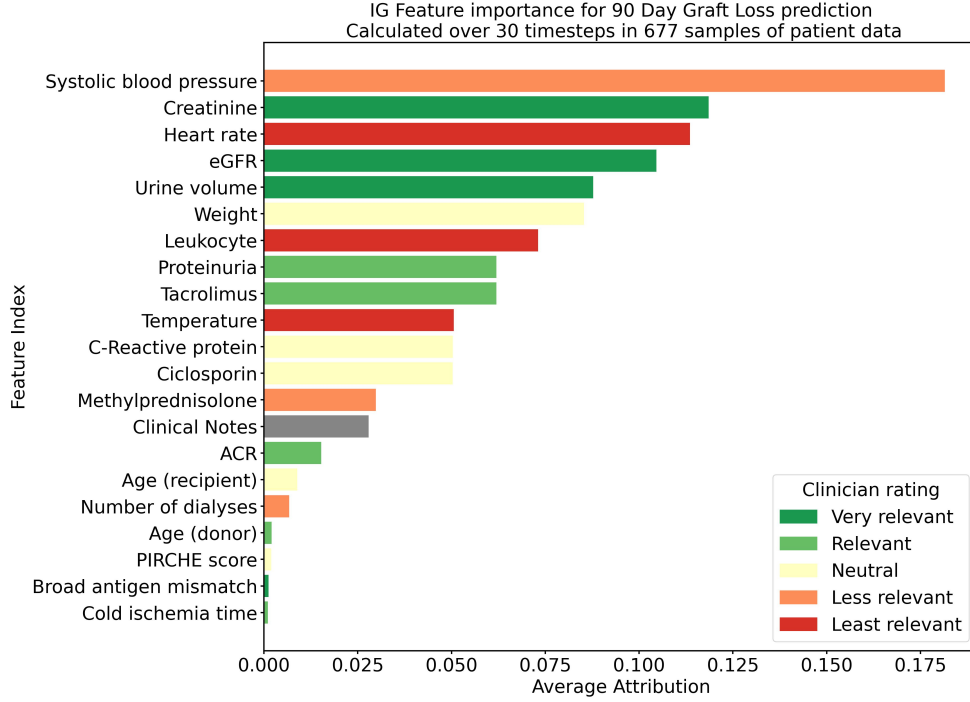


Figure 9: Mean absolute IG values for each feature, indicating their relative importance in TFN predictions for graft loss. The relevance rating computed from our clinical expert study is represented by colour. Although the less relevant feature systolic blood pressure appears as the top feature, kidney function markers such as creatinine and eGFR were identified by the study as top-three features and appear among the top four features in the IG-based attributions.

Increasing the prediction horizon from 30 to 240 timesteps yielded only marginal improvements in NDCG for both IG and TIMING, whilst substantially increasing runtime. For instance, TIMING’s NDCG improved by just 0.009 over this range. Consequently, a 30-step horizon was adopted for all subsequent experiments.

Beyond feature attributions, we analysed the relevance of individual timesteps. Figure 11 shows feature $\times$ timestep attributions as a heatmap, revealing that the model’s decisions are almost entirely driven by the last few timesteps, with earlier observations contributing negligibly. This pattern persisted regardless of horizon size. Applying CRP to the output layer confirmed this finding: the final neuron of the TM-LSTM — responsible for processing the last timestep — consistently exhibited the highest relevance, with multiple cases reaching 100%, indicating that it was the sole contributor to the model output.

A case in which TIMING outperformed SHAP was observed for the 30-day mortality attributions. Here, TIMING achieved an NDCG score of 0.98. As shown in Figure 12, TIMING indicates that TFN places strong emphasis on

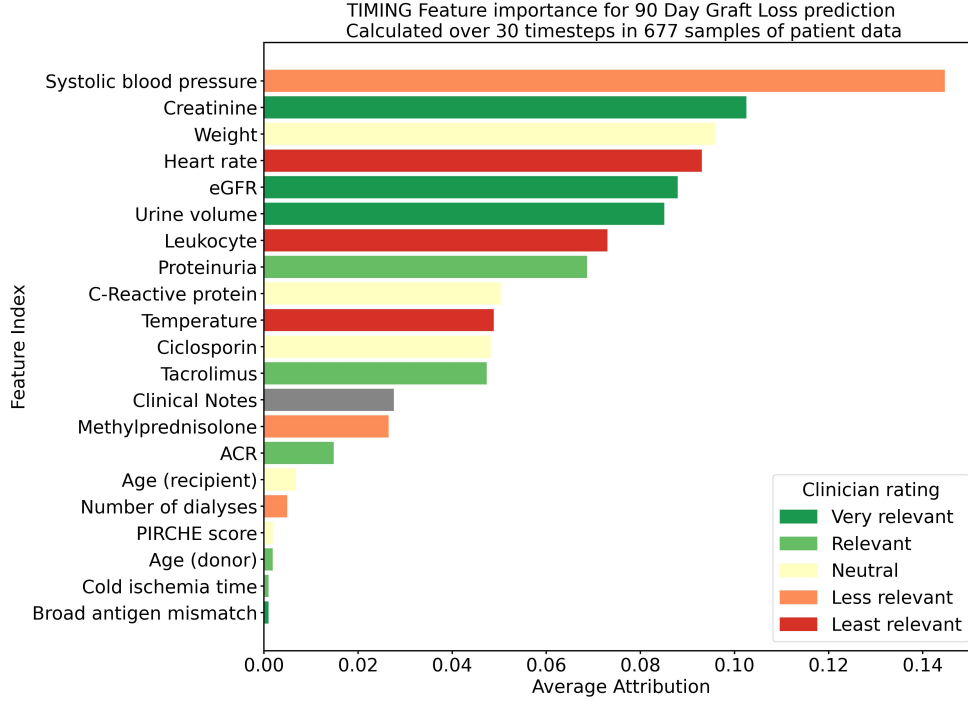


Figure 10: Mean absolute TIMING values for each feature, indicating their relative importance in TFN predictions for graft loss. The relevance rating computed from our clinical expert study is represented by colour. Although the less relevant feature systolic blood pressure appears as the top feature, kidney function markers such as creatinine and eGFR were identified by the study as top-three features and appear among the top five features in the TIMING-based attributions.

temperature and heart rate, which were previously missed by SHAP.

6.2 Uncertainty analysis

Because TFN was shown to be overconfident in its probability estimates, often predicting too high a likelihood of an event occurring [3], we performed uncertainty quantification to determine whether this behaviour might stem from the TM-LSTM having difficulty handling uncertainty.

To do so, we performed 90-day graft loss estimation. The full test set was split in half: the first half served as a calibration set for ACI and RCPS, which were then used to estimate the coverage of individual temporal features on the second half.

As shown in Figure 13, both methods achieved strong performance, with TFN reaching a mean coverage of over 90% across all features. ACI yielded slightly higher coverage whilst simultaneously producing narrower prediction intervals, as shown in Figure 14, suggesting greater precision. RCPS, in turn,

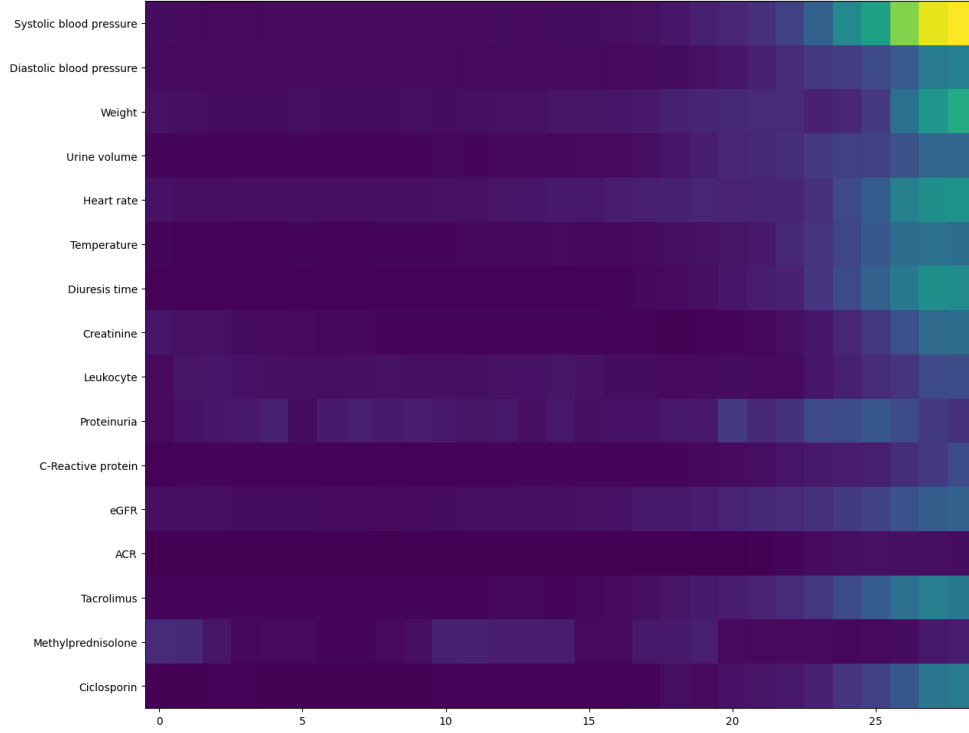


Figure 11: Mean absolute IG values for each feature $\times$ timestep pairing, indicating their relative importance in TFN predictions for graft loss. The model shows a strong focus on recent events while almost ignoring the early to mid patient trajectory.

achieved a nearly 20-times faster runtime.

7 Discussion

7.1 Explainability Analysis

The strong NDCG scores achieved by IG and TIMING (0.85) and SHAP (0.96) for graft loss prediction suggest that TFN’s learned representations are broadly consistent with clinical reasoning. The decline in SHAP’s Spearman correlation between the two studies ($\rho = 0.62$ to $\rho = 0.34$) is likely an artefact of the restructured study design: the updated questionnaire produced a more balanced distribution of relevance ratings, with a greater proportion of features rated as less than *Relevant*, whereas the previous study marked only 8 of 26 features below that threshold. This redistribution makes rank correlation a stricter measure, even when the underlying agreement on the most important features remains intact.

The concentration of model relevance in the final timesteps is a direct consequence of the TM-LSTM’s temporal decay mechanism (Equation 3). Whilst

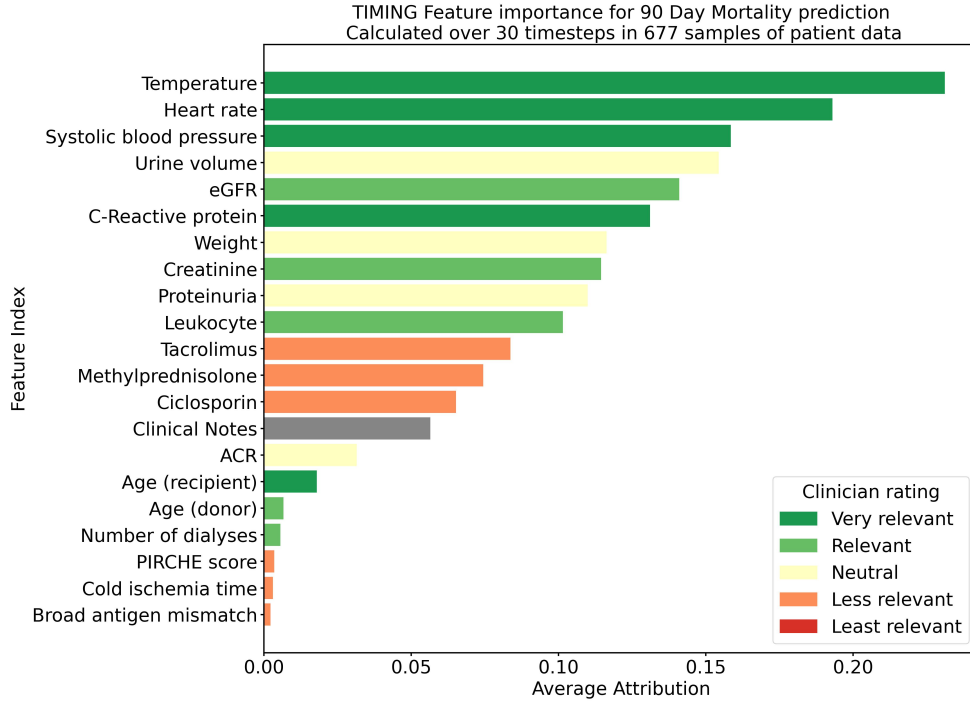


Figure 12: Mean absolute TIMING values for each feature, indicating their relative importance in TFN predictions for mortality. The relevance rating computed from our clinical expert study is represented by colour. This is a strong result, featuring a near-perfect ranking, although it still misses the important relationship between mortality and patient age.

this behaviour is architecturally expected, it appears overly pronounced in practice, frequently causing the model to disregard earlier events in the patient trajectory entirely. This also likely explains why TIMING does not significantly outperform classic IG for TFN: if temporal relationships decay to negligible levels already within the first layer, there is little signal for TIMING to capture significant, long-term temporal patterns. The negligible improvement in NDCG with increasing horizon further supports this interpretation — a longer horizon provides no benefit if the model itself ignores the additional context.

The high NDCG for mortality prediction again highlights TFN’s focus on the most recent clinical observations. In the final days before death, features such as body temperature and heart rate would naturally undergo pronounced physiological changes irrespective of the underlying cause, making them genuinely predictive from the model’s perspective. However, TIMING also missed the importance of patient age, which SHAP had previously captured perfectly as the top feature in mortality prediction. The reverse case, namely overestimating the importance of a feature, also requires careful interpretation rather than blind reliance on the study results. Across all tasks, clinicians were asked

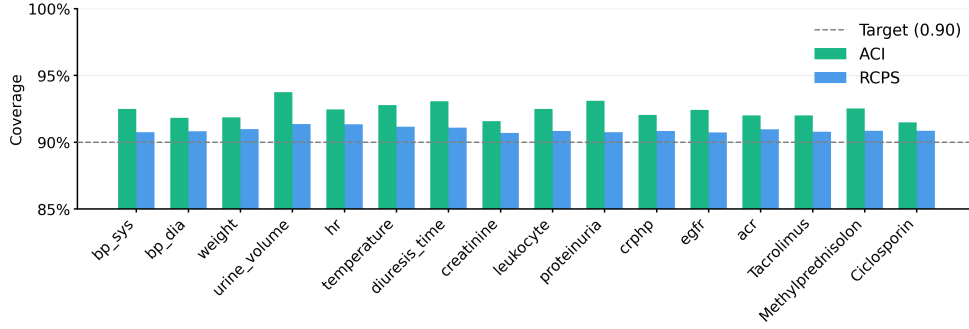


Figure 13: Coverage per feature, with the threshold $\alpha = 0.9$ shown as a dotted line. Coverage of 90% indicates that the model predicts this feature within the prediction interval 90% of the time. When the threshold is reached or exceeded, the model may be considered not to exhibit an uncertainty problem on average.

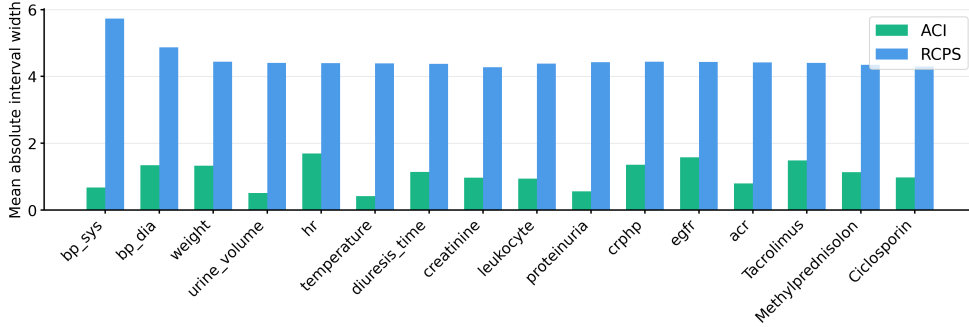


Figure 14: Mean absolute interval width per feature for both ACI and RCPS. A wide interval can indicate low confidence, although interval width also depends on the feature, as a larger value range naturally leads to a wider interval. ACI shows greater variance in its average widths, but overall much narrower intervals than RCPS.

to assess feature relevance specifically in the context of an event occurring *as a consequence of kidney transplantation*. In the case of systolic blood pressure for the graft loss task, for example, the feature was rated as less relevant by the expert panel, yet elevated systolic pressure has previously been associated with increased mortality risk in patients with chronic kidney disease because it places the kidneys under considerable haemodynamic stress [41]. Although the cited paper found no correlation between systolic blood pressure and patients eventually requiring KTx, the assumption that a weakened transplanted organ may be at risk of failure during periods of markedly increased stress appears plausible.

7.2 Uncertainty Analysis

The expected areas of underconfidence in TFN were not observed; instead, the model achieved strong coverage in all cases under both metrics. On a per-patient basis, isolated coverage failures were observed, but in almost all cases these could be traced back to data quality issues in the individual patient trajectory, such as missing measurements for the affected feature. As already noted in Section 6.2, ACI achieved much narrower intervals, with a sensible variance in their width. Considering the realistic range of values for, for example, temperature and heart rate, it is reasonable that the greater variance in heart rate also leads to wider intervals on average. RCPS, by contrast, showed almost no variance in interval width per feature and similarly little variation in coverage, thereby producing more *stable* results.

7.3 Future work

Future research should focus on the problems associated with the TM-LSTM decay function mentioned above. Achieving fuller utilisation of the patient history without losing focus on recent events should lead to better model performance and greater interpretability, by producing more informative time-step attributions and, hopefully, more interesting results from TIMING.

One way to achieve such utilisation would be to retrain TFN with this specific objective in mind, as the strength of the decay function is set during training and the problem may simply stem from the decay being too strong in its current form. Another promising approach lies in so-called *State Space Models* such as Mamba [42], a relatively new architecture that has recently shown strong performance in sequence modelling. Exploring the possibility of replacing the current TM-LSTM with an SSM would therefore be of considerable interest. In addition, IG and TIMING attributions would likely improve if a TFN-specific baseline were constructed [43]. This work focused on attributions for temporal data, but all provided local explainability models are also able to provide attributions for a patient’s clinical notes. At present, these attributions are not used, because they indicate only which *embedding* was relevant, and since Medgte-hybrid is only capable of encoding, there is currently no way to determine what an embedding corresponds to. Future work could therefore attempt to set up a decoder model specific to the architecture, so that attributions on the actual notes could be obtained.

Originally, we intended to do substantially more with CRP. In practice, however, this task proved rather difficult and consumed a great deal of time while yielding few results. The main drawback of CRP is that it requires prior knowledge of what to search for and where to search for it. To obtain such knowledge, the *zennit-crp* library also provides RelMax, as well as the possibility of generating an attribution graph of a model. In practice, however, while CRP already

required several days of work to restructure TFN correctly so that it became compatible with the library, the additional effort required to make RelMax work was ultimately considered beyond the project scope. Zennit-crp has a clear focus on image data, and applying RelMax would have required substantial refactoring to fit TFN while preserving compatibility with CRP. The use of the attribution graph was also discarded, because multiple attempts failed after days of trying to compute it. Future work could return to this idea, as RelMax would likely again enable a better understanding of TFN. One possible use case would be to use RelMax to identify patient subgroups through concepts such as high or low creatinine, and then assess whether these subgroups already form clusters in the embedding space.

After integrating so many different attribution techniques, it may be worthwhile to pursue a strategy similar to that proposed by [4]. They introduced the *XAI Orchestrator*, an LLM-based model built specifically to receive the outputs of multiple attribution techniques and help physicians obtain a faster and more nuanced interpretation of the model by analysing these results jointly.

Because both ACI and RCPS showed no areas of underconfidence in TFN, a planned direct model adaptation was omitted. Instead, adapting the model with case-specific rules would probably be more interesting. One possibility would be to compute the risk of a decision in an uncertain case through RCPS and then allow the model to output a warning instead of a prediction if the risk exceeded a certain threshold, thereby supporting real-world use. Alternatively, especially in research settings, an incorrect prediction could be used as feedback to adapt the model, with the risk of that prediction serving as a factor controlling the strength of the adaptation. In an effort to achieve even better conformal prediction results, ACI could be replaced by the more complex CPTC [27] algorithm. Specifically engineered for time-series data with many change points, it would likely be a strong fit for medical data. However, since ACI is already showing strong performance, this would be the lowest-priority future work outlined in this section.

8 Conclusion

This thesis investigated the interpretability and reliability of machine learning models in the context of multimodal approaches for medical use cases. By showing how multiple attribution techniques can be applied at both the local and global levels to TFN, this work made the model more understandable, provided new insights that further validate its strong performance, and opened up new directions for research.

Through the use of different techniques, we identified several discrepancies with clinical opinion, but each of these remained medically justifiable in the given context. We showed that misalignment should be treated as an oppor-

tunity for analysis rather than as a strict error. We also found that, while the decay function was probably a key factor in TFN’s success, it may also be one of its main limitations. At the very least, we showed that it constrains interpretability.

We also demonstrated multiple ways to quantify TFN’s predictive uncertainty and showed that the model performed with a high degree of confidence, thereby strengthening reliability and reinforcing the foundation for trust. In addition, we outlined how the proposed metrics could be used in future work to adapt the model to situations that are more complex than the average case.

Finally, this work contributes the results of a novel expert study in which multiple nephrologists provided critical insight into the factors they prioritise when making decisions in the context of three challenging clinical events. These results are likely to prove valuable for future research.

In conclusion, AURA provides a template for the comprehensive evaluation of medical AI systems in the KTx context and, if such systems meet these standards, helps bring them closer to real clinical deployment.

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Appendix

Feature	Graft Loss	Rejection	Mortality
ACR	70.82	60.35	46.38
Age (donor)	59.12	53.50	37.69
Age (recipient)	59.12	65.05	91.75
Blood group (donor)	31.76	46.85	12.69
Blood group (recipient)	33.65	53.45	21.06
Blood pressure	33.71	26.90	85.56
Broad antigen mismatch	76.65	91.75	35.81
C-Reactive protein	39.65	35.40	75.25
Ciclosporin	41.65	47.05	29.75
Cold ischemia time	63.06	71.85	39.88
Creatinine	82.53	75.20	64.94
eGFR	82.53	65.30	62.88
Gender (donor)	25.76	35.05	14.75
Gender (recipient)	21.88	51.70	41.94
Heart rate	14.00	15.25	71.13
Leukocyte	27.94	28.70	66.94
Methylprednisolone	31.88	48.75	40.13
Number of dialyses	31.76	27.05	52.50
PIRCHE score	49.35	56.95	27.50
Proteinuria	67.00	68.60	46.38
Tacrolimus	63.06	70.20	40.13
Temperature	25.88	33.70	71.13
Type of donation	70.88	71.85	48.25
Urine volume	80.59	75.15	48.31
Weight	47.47	35.25	50.31

Table 2: Clinical feature relevance scores across three tasks, computed based of the clinician answers from the performed study. The results show the age of the patient as most important for mortality, a broad antigen mismatch as most relevant for rejection, and creatinine and eGFR as equally most important for the estimation of graft loss.

Feature	Score	Clinical Rating
eGFR	82.53	Very relevant
Creatinine	82.53	Very relevant
Urine volume	80.59	Very relevant
Broad antigen mismatch	76.65	Very relevant
Type of donation	70.88	Relevant
ACR	70.82	Relevant
Proteinuria	67.00	Relevant
Tacrolimus	63.06	Relevant
Cold ischemia time	63.06	Relevant
Age (donor)	59.12	Relevant
Age (recipient)	59.12	Relevant
PIRCHE score	49.35	Neutral
Weight	47.47	Neutral
Ciclosporin	41.65	Neutral
C-Reactive protein	39.65	Neutral
Blood pressure	33.71	Less relevant
Blood group (recipient)	33.65	Less relevant
Methylprednisolone	31.88	Less relevant
Number of dialyses	31.76	Less relevant
Blood group (donor)	31.76	Less relevant
Leukocyte	27.94	Least relevant
Temperature	25.88	Least relevant
Gender (donor)	25.76	Least relevant
Gender (recipient)	21.88	Least relevant
Heart rate	14.00	Least relevant

Table 3: Clinical feature relevance ratings for graft loss, sorted by score. eGFR, Creatinine, Urine volume and Broad antigen mismatch were identified as the most relevant features.

Feature	Score	Clinical Rating
Broad antigen mismatch	91.75	Very relevant
Creatinine	75.20	Very relevant
Urine volume	75.15	Very relevant
Cold ischemia time	71.85	Very relevant
Type of donation	71.85	Very relevant
Tacrolimus	70.20	Very relevant
Proteinuria	68.60	Very relevant
eGFR	65.30	Relevant
Age (recipient)	65.05	Relevant
ACR	60.35	Neutral
PIRCHE score	56.95	Neutral
Age (donor)	53.50	Neutral
Blood group (recipient)	53.45	Neutral
Gender (recipient)	51.70	Neutral
Methylprednisolone	48.75	Neutral
Ciclosporin	47.05	Neutral
Blood group (donor)	46.85	Neutral
C-Reactive protein	35.40	Less relevant
Weight	35.25	Less relevant
Gender (donor)	35.05	Less relevant
Temperature	33.70	Less relevant
Leukocyte	28.70	Least relevant
Number of dialyses	27.05	Least relevant
Blood pressure	26.90	Least relevant
Heart rate	15.25	Least relevant

Table 4: Clinical feature relevance ratings for graft rejection, sorted by score. Broad antigen mismatch is identified as the most relevant feature, with multiple other also being very relevant.

Feature	Score	Clinical Rating
Age (recipient)	91.75	Very relevant
Blood pressure	85.56	Very relevant
C-Reactive protein	75.25	Very relevant
Temperature	71.13	Very relevant
Heart rate	71.13	Very relevant
Leukocyte	66.94	Relevant
Creatinine	64.94	Relevant
eGFR	62.88	Relevant
Number of dialyses	52.50	Relevant
Weight	50.31	Neutral
Urine volume	48.31	Neutral
Type of donation	48.25	Neutral
ACR	46.38	Neutral
Proteinuria	46.38	Neutral
Gender (recipient)	41.94	Less relevant
Methylprednisolone	40.13	Less relevant
Tacrolimus	40.13	Less relevant
Cold ischemia time	39.88	Less relevant
Age (donor)	37.69	Less relevant
Broad antigen mismatch	35.81	Less relevant
Ciclosporin	29.75	Less relevant
PIRCHE score	27.50	Less relevant
Blood group (recipient)	21.06	Least relevant
Gender (donor)	14.75	Least relevant
Blood group (donor)	12.69	Least relevant

Table 5: Clinical feature relevance ratings for patient mortality, sorted by score. Patient age and blood pressure being the most relevant features.